MATHEMATICAL FOUNDATIONS FOR AI AND DATA SCIENCE

FAQ — UNIT 1 MOST-ASKED / REPEATED PYQs

LEVEL 1 — DIRECT / FUNDAMENTAL

1. [Systems / Gaussian elimination] Use Gaussian Elimination to solve: a + b + c = 6, a + 2b + 2c = 11, and 2a + 3b - 4c = 3.
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

2. [Systems / Gaussian elimination] Use Gaussian Elimination to decide whether the planes 6x - 3y + 3z = 2, 2x - y + z = 1, and 3x + 2y - 4z = 4 have a common point in R^3.
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

3. [Systems / Gaussian elimination] For the same plane system, determine what happens if the right-hand side of the second equation is changed to 2/3 instead of 1.
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

4. [Gauss-Jordan inverse] Use the Gauss-Jordan method to invert A = [[2, 1, 0], [-1, 2, 1], [0, -1, 2]].
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019,2025

5. [Linear independence] Determine whether the vectors x = (1, 3, 2) and y = (2, 6, 4) are linearly independent or not.
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

6. [Fundamental subspaces] Find the basis and dimension of C(A), N(A), and N(A^T) for the given matrix A.
Source year: 2017; Repeated theme years: 2016, 2017, 2018, 2019

7. [Linear transformations] Find the Range and Kernel of the transformations: (i) T(v1, v2) = (v2, v1), (ii) T(v1, v2, v3) = (v1, v2), and (iii) T(v1, v2) = (v1, v1).
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

LEVEL 2 — APPLICATION

8. [Gaussian elimination] Apply Gaussian Elimination to 2x - 3y + 4z = 7, x - y + 2z = 10, and 8x - 11y + 16z = 41. Decide whether the system is consistent and solve it if it is. What happens if the coefficient of z in the second equation is replaced by 3?
Source year: 2017; Repeated theme years: 2016, 2017, 2018, 2019,2024

9. [LU decomposition] Compute L and U for A = [[a, a, a, a], [a, b, b, b], [a, b, c, c], [a, b, c, d]] and find the conditions on a, b, c, d for A = LU with four pivots.
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019,2026

10. [LU decomposition / rank] Find L and U for A = [[2, 4, 1, 5, 2], [4, 5, 3, 8, 1], [2, 5, 4, 1, 8], [6, 6, 7, 3, 1]]. What is the rank of A?
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

11. [Column space / special solutions] If C(A) is spanned by (1, 3, -5), (0, 5, c), and (-2, -1, 0), find c for which C(A) is a plane in R^3 and for which it is all of R^3.
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

12. [Column space / special solutions] Find the special solutions to Ax = 0 when A has the preceding vectors as columns, using the value of c for the plane case.

Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

13. [Span / null space] For what value of lambda will (1, 3, -5), (0, 5, lambda), and (-2, -1, 0) span a two-dimensional subspace? For this value of lambda, find bases for C(A) and N(A^T).
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

14. [Linear independence] If a, b, c are linearly independent vectors, determine whether the vectors a - b, b - c, and c - a are also linearly independent.
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

LEVEL 3 — HIGH-VALUE / REPEATED

15. [Four fundamental subspaces] Find the four fundamental subspaces for A = [[1, 3, 3, 2], [2, 6, 9, 7], [-1, -3, 3, 4]].
Source year: 2016; Repeated theme years: 2016, 2017, 2018, 2019

16. [Column space / null space] Find matrix A such that the given set is C(A): {(b - c, 2b + c + d, 5c - 4d, d)^T : b, c, d are real}. Hence find the solutions of Ax = 0 and A^T y = 0.
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

17. [Linear transformations] Let T: R^3 -> R^3 be defined by T(x, y, z) = (x - y + z + t, x + y + 2z - t, x + y + 3z - 3t). Find the basis and dimension of Range(T).
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019,2024

18. [Linear transformations] For the same transformation T, find the basis and dimension of Kernel(T).
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

19. [Linear transformations] Determine the Kernel and Range of T([x, y, z]^T) = [x + 2y - z, y + z, x + y - 2z]^T.
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

20. [Linear transformations] For the preceding transformation, find the dimensions of the null space and column space of the matrix of T.
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019,2022

21. [Left null space] If C(A) is spanned by (0, 1, 1), (1, 2, 3), (4, 5, 6), and (7, 8, 9), find all vectors that span the left null space of A.
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

22. [Left null space] For the preceding A, determine whether b = (4, -2, 2) belongs to the left null space.
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

23. [Left null space / dimensions] For the preceding A, find the dimensions of the row space and left null space.
Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

24. [Row space / rank] If the row space is spanned by (2, 4, 6, 4), (2, 5, 7, 6), and (2, 3, 5, c), find c for which C(A) is a plane in R^3 or all of R^3.
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

25. [Row space / null space] For the plane case in the preceding question, identify the free variables and write the special solutions of Ax = 0.
Source year: 2019; Repeated theme years: 2016, 2017, 2018, 2019

26. [Vector spaces] Find the basis and dimension of the space of all vectors in R^n whose components add to zero.

Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019

27. [Vector spaces] Find the basis and dimension of the space of all vectors in R^n whose components are equal.

Source year: 2018; Repeated theme years: 2016, 2017, 2018, 2019